

Empirical Analysis of Phase-Driven Humanoid Locomotion: Diffeomorphic Trajectory Regularization versus Reflex-Induced Spectral Jerk and Parasitic Work

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Abstract—Central Pattern Generators (CPGs) are widely celebrated for synthesizing smooth, continuous kinematic manifolds that bypass the non-differentiable spatial transitions of classical discrete Finite-State Machines (FSMs). However, the underlying mechanical, spectral, and thermodynamic consequences of coupling active vestibulospinal posture reflexes directly onto these continuous phase manifolds remain poorly quantified. This paper presents an empirical, multi-sensor comparative deconstruction of both locomotion paradigms using an underactuated 12-DOF anthropomorphic biped simulated within the MuJoCo physics engine. While high-frequency kinematic telemetry confirms that diffeomorphic phase mapping successfully regularizes the swing-foot Cartesian limit cycle into a twice-differentiable (C^2) manifold, frequency-domain audits reveal a critical liability: a two-order-of-magnitude high-frequency actuator jerk penalty injected by unfiltered inertial attitude corrections. Furthermore, a thermodynamic stance-phase analysis demonstrates that naive phase-driven ankle compliance accelerates downward Center of Mass yielding, inflating late-stance parasitic negative braking work by over 54% compared to rigid spatial tracking. By cross-examining these synchronized spatiotemporal and energetic data streams, this work provides a rigorous empirical foundation for establishing impedance-scheduled reflex boundaries in underactuated humanoid robots.

Index Terms—Bipedal Locomotion, Central Pattern Generators, Finite State Machines, Diffeomorphic Manifolds, Cost of Transport, Spectral Jerk, Humanoid Robotics.

I. INTRODUCTION

Developing robust, autonomous bipedal locomotion remains one of the most formidable challenges in humanoid robotics. The inherent physical dynamics of legged systems—characterized by high-dimensional non-linear state trajectories, inherently unstable inverted-pendulum mechanics, and hybrid, unilateral ground-contact transitions—make analytical gait synthesis exceptionally difficult. To safely debug

joint-level control laws and evaluate multi-point contact states prior to physical hardware deployment, modern robotics relies heavily on high-fidelity, rigid-body physics engines such as MuJoCo [3]. Within these simulated testing environments, the core engineering objective is authoring lower-limb joint trajectories that achieve both dynamic stability and energetic efficiency.

Historically, locomotive gait generation has been dominated by discrete Finite-State Machines (FSMs). In a classical FSM architecture, the bipedal stepping cycle is partitioned into distinct spatial phases—such as single-leg support, forward leg swing, heel-strike impact, and double-support weight transfer [1]. The joint-level actuators track a sequence of spatial keyframe dictionaries using explicit temporal interpolators. While computationally lightweight and intuitive to construct, pure spatial keyframing enforces a rigid, unyielding mapping between the temporal domain and Cartesian coordinates. This structural rigidity forces non-differentiable velocity transitions at state boundaries, resulting in harsh, mechanical impacts during foot touchdown and severely restricting the biped's stable velocity upper bound.

To overcome the geometric brittleness of discrete keyframing, recent literature heavily champions continuous Central Pattern Generators (CPGs). Inspired by biological neural networks within the vertebrate spinal cord, continuous CPG architectures replace the totalitarian clock of an FSM with an autonomous, monotonically progressing phase manifold $\Phi \in [0, 1)$ [6]. By driving the Cartesian joint spaces through coupled, twice-differentiable (C^2) splines indexed to this phase variable, CPGs smoothly regularize stepping geometry. This spatial regularization eliminates harsh vertices, allowing humanoids to synthesize fluid, biomimetic Cartesian limit

cycles while modulating stepping frequency and stride length independently [4].

However, an open-loop kinematic trajectory alone cannot stabilize a floating-base humanoid across non-planar or micro-perturbed environments. To maintain Center of Mass (CoM) attitude, the low-level motor torques must continuously incorporate live sensor feedback—specifically high-frequency attitude and angular velocity streams from an onboard Inertial Measurement Unit (IMU)—emulating the biological vestibulospinal reflex [7]. While existing control literature exhaustively documents the geometric superiority of continuous CPGs over discrete FSMs, a critical vulnerability remains largely unaddressed: the hidden mechanical fatigue and thermodynamic penalties incurred when naively superimposing active, high-frequency IMU reflexes directly onto compliant phase manifolds.

This paper introduces an empirical diagnostic framework that deconstructs this precise control tension. By treating an underactuated 12-DOF anthropomorphic biped as a black-box data generator within MuJoCo, we implement both a classical discrete FSM keyframe interpolator and a continuous diffeomorphic CPG. We extract and analyze synchronized 100 Hz streams of Cartesian base translations, vestibular attitude, actual motor torques, and joint velocities. By cross-examining the resulting exhaust telemetry, we quantitatively demonstrate that while phase manifolds successfully regularize stepping geometry, they introduce profound mechanical and energetic liabilities when coupled with un-filtered posture reflexes.

The primary contributions of this work are threefold:

- The mathematical formulation and MuJoCo simulation of a comparative bipedal control architecture contrasting discrete spatial keyframes against continuous diffeomorphic phase manifolds.
- An empirical frequency-domain deconstruction proving that superimposing raw 100 Hz IMU attitude corrections onto continuous CPG manifolds injects severe high-frequency actuator jerk (10^7 PSD) into the physical structure compared to the inert FSM baseline (10^4 PSD).
- A thermodynamic stance-phase audit demonstrating that naive phase-driven ankle compliance accelerates Center of Mass yielding, inflating late-stance parasitic negative braking work by over 54% compared to rigid spatial tracking.

II. REVIEW OF RELATED LITERATURE

The quest to build stable, autonomous bipedal robots has historically progressed along two fundamentally different research philosophies: strict mathematical trajectory optimization versus biologically-inspired neural compliance. Understanding how these two paradigms attempt to solve the “falling over” problem is essential for contextualizing why our empirical findings matter.

A. Trajectory Optimization versus Continuous Manifolds

Classical legged robotics heavily favors deterministic, time-indexed control layers. Model Predictive Control (MPC) is

perhaps the most famous standard in the industry, operating by calculating optimal joint angles and ground reaction forces over a short, rolling time horizon [3]. While MPC offers brilliant mathematical guarantees that a robot will not fall over under ideal conditions, it requires massive real-time computational power. More importantly, purely time-indexed formulations are inherently “brittle”; if the physical robot accidentally steps on a rock and hits the ground 0.1 seconds earlier than the internal clock expected, the tracking error spikes and the system often fails.

To escape the strict, unforgiving nature of stopwatches, early pioneering work by Ijspeert et al. demonstrated the immense value of continuous Central Pattern Generators [6]. Instead of using time, CPGs map locomotion to autonomous, repeating phase oscillators. This breakthrough allowed robots to smoothly speed up or slow down their gait dynamically. Ames et al. later formalized this concept into “Phase Variables” using Hybrid Zero Dynamics (HZD), completely decoupling the robot’s posture from a clock and tying it directly to the forward spatial movement of the stance leg [4]. This continuous phase mapping successfully solved the geometric “brittleness” problem, creating beautiful, fluid steps.

B. Biological Compliance and Postural Reflexes

However, creating a beautiful geometric step in a simulator is only half the battle. Real-world biological walking is incredibly messy, relying heavily on dynamic physical compliance. For example, humans naturally relax their trailing ankle as it leaves the ground to allow the foot to passively roll forward, while aggressively stiffening their lead knee just before heel-strike to absorb the impact [1].

To capture this biological efficiency, modern control researchers have started building Variable Impedance Scheduling into their robots. Recent advances in deep reinforcement learning (RL) have produced “rapid motor adaptation” loops. These AI-driven policies allow a humanoid to sense sudden changes in motor current and instantly adjust how stiff or compliant its joints are, much like a human tensing up on an icy sidewalk [2], [7].

Alongside these compliance schedules, almost all practical walking robots run a local “vestibulospinal reflex” in the background—a simple control loop that reads the robot’s IMU and twitches the ankles or hips to keep the torso perfectly upright [5].

While these compliant, reflex-driven architectures sound incredibly robust in theory, they are rarely subjected to rigorous, frequency-domain audits. As our results will demonstrate, when you take an active, high-frequency IMU reflex and blindly plug it into a soft, compliant continuous phase manifold, the underlying motors end up paying a massive, hidden tax.

III. MATHEMATICAL FORMULATION AND CONTROL ARCHITECTURE

To evaluate the systemic trade-offs between discrete and continuous gait synthesis, we model the anthropomorphic

biped as a floating-base kinematic tree comprising 12 actuated degrees of freedom (DOF) and an unactuated 6-DOF base. Let the full configuration space be defined as $\mathbf{q} \in \mathbb{R}^{18}$, such that $\mathbf{q} = [\mathbf{q}_{\text{base}}^T, \mathbf{q}_{\text{joints}}^T]^T$, where $\mathbf{q}_{\text{joints}} \in \mathbb{R}^{12}$ represents the internal rotational joint angles.

A. Diffeomorphic Trajectory Regularization

The classical FSM baseline enforces a discrete spatial trajectory by linearly interpolating joint postures across a finite set of keyframe dictionaries $\mathcal{K} = \{\mathbf{k}_0, \mathbf{k}_1, \dots, \mathbf{k}_m\}$. The joint reference trajectory $\mathbf{q}_{\text{ref}}(t)$ is parameterized by a normalized temporal accumulator $\alpha(t) \in [0, 1]$ driven by a fixed timer:

$$\alpha(t) = 3 \left(\frac{t}{\Delta T} \right)^2 - 2 \left(\frac{t}{\Delta T} \right)^3 \quad (1)$$

where ΔT is the hardcoded duration of the state. This formulation inherently maps time to configuration space, yielding non-differentiable spatial vertices at state transitions where joint accelerations spike.

Conversely, the proposed continuous CPG decouples the temporal domain from spatial execution by defining a monotonic phase manifold $\Phi = [0, 1]$. The phase variable progresses linearly according to the commanded step frequency:

$$\dot{\phi} = \frac{1}{T_{\text{swing}}} \quad (2)$$

To synthesize the swing-foot Cartesian limit cycle, we establish a diffeomorphic mapping $\psi : \Phi \rightarrow \mathbb{R}^3$ that regularizes the non-differentiable vertices into a smooth, C^2 -continuous teardrop manifold in the sagittal plane (x - z):

$$x_{\text{swing}}(\phi) = x_{\text{stance}} + S_{\text{scale}} \cdot \gamma_x(\phi) \quad (3)$$

$$z_{\text{swing}}(\phi) = z_{\text{nominal}} + H_{\text{peak}} \cdot \gamma_z(\phi) \quad (4)$$

where S_{scale} and H_{peak} are dynamically coupled amplitude parameters scaled linearly by the velocity demand, and $\gamma(\phi)$ represents a twice-differentiable cubic spline basis. Because $\mathbf{q}_{\text{ref}}(\phi)$ is mapped continuously via local inverse kinematics, spatial continuity is preserved across the entire stepping cycle.

B. Quantification of Reflex-Induced Spectral Jerk

To mitigate vestibular drift and prevent tipping, both controllers utilize a low-overhead vestibulospinal postural reflex. The instantaneous inertial attitude error is mapped directly to a compensatory ankle pitch torque reference:

$$\Delta \tau_{\text{ankle}} = -K_p \theta_{\text{pitch}} - K_d \dot{\theta}_{\text{pitch}} \quad (5)$$

where θ_{pitch} and $\dot{\theta}_{\text{pitch}}$ are the raw 100 Hz attitude and angular velocity feedback streams harvested directly from the base IMU.

However, injecting unfiltered high-frequency sensor noise directly into a continuous phase manifold induces severe mechanical stress. To quantify this liability, we analyze the discrete time-derivative of the actual motor torques, defined as Actuator Jerk $\mathbf{j}_{\tau}[n] \in \mathbb{R}^{12}$:

$$\mathbf{j}_{\tau}[n] = \frac{\boldsymbol{\tau}[n] - \boldsymbol{\tau}[n-1]}{\Delta t} \quad (6)$$

where $\Delta t = 0.01$ s is the sampling period. We apply a Fast Fourier Transform (FFT) over an N -sample window to calculate the Power Spectral Density (PSD) of the actuator jerk, $S_{jj}(f_k)$, at discrete frequency bins f_k :

$$S_{jj}(f_k) = \frac{1}{N \Delta t} \left| \sum_{n=0}^{N-1} j_{\tau,i}[n] e^{-i \frac{2\pi}{N} kn} \right|^2 \quad (7)$$

C. Thermodynamic Exhaust and Parasitic Work

To evaluate locomotion efficiency, we analyze the instantaneous thermodynamic power across all 12 actuators. Let the mechanical power of joint i at time t be defined as $P_i(t) = \tau_i(t) \cdot \dot{q}_i(t)$.

During the late stance phase, a rigid position controller actively opposes the biped's forward momentum to track a static dictionary pose, resulting in negative mechanical work. To capture this energetic liability, we isolate the total **Parasitic Work** ($W_{\text{parasitic}}$) dissipated at the trailing ankle during the stance window $\mathcal{T}_{\text{stance}} = [t_{\text{touchdown}}, t_{\text{toe-off}}]$:

$$W_{\text{parasitic}} = \int_{t_{\text{touchdown}}}^{t_{\text{toe-off}}} \min(0, P_{\text{ankle}}(t)) dt \quad (8)$$

Furthermore, we define the global thermodynamic efficiency across the complete trajectory using the absolute Cost of Transport (CoT):

$$\text{CoT} = \frac{1}{M g d_{\text{total}}} \int_0^{T_{\text{total}}} \sum_{i=1}^{12} |\tau_i(t) \cdot \dot{q}_i(t)| dt \quad (9)$$

where $M = 35.4$ kg is the system mass, g is gravity, and d_{total} is net Cartesian displacement over the simulation horizon.

IV. EXPERIMENTAL SETUP

A. System Architecture and Data Acquisition

The experimental framework evaluates our locomotion control architecture using a custom-designed, underactuated 12-DOF anthropomorphic biped. The simulated robotic platform is architected with 12 active rotational degrees of freedom (DoFs) distributed symmetrically across the lower extremities. These active joints map directly to the anatomical equivalents of hip yaw, hip roll, hip pitch, knee pitch, ankle pitch, and ankle roll for both the left and right legs. Furthermore, the spatial state vector of the unactuated floating-base pelvis is explicitly defined across 6 DoFs to isolate and track sagittal, lateral, and vertical translations, alongside roll, pitch, and yaw rotational trajectories. Actuator torque limits and joint range-of-motion boundaries are mathematically constrained within the model to mirror realistic electromechanical hardware specifications.

To establish a highly accurate and authentic biological tracking baseline, empirical continuous locomotion datasets were recorded from healthy human subjects. Kinematic telemetry was captured utilizing a network of WitMotion WT901 9-axis inertial measurement units (IMUs). Each WT901 sensor node is equipped with a high-performance microprocessor integrating a 3-axis accelerometer, a 3-axis gyroscope, and a 3-axis magnetometer. To minimize integration drift and external

magnetic interference, the sensors natively compute spatial orientation utilizing an onboard dynamic Kalman filtering algorithm.

To ensure the integrity of the biomechanical data, the IMU sensors were securely affixed to the subjects using custom elastic strapping on the anterior mid-thighs and lateral shanks, deliberately positioned to minimize soft-tissue artifacts (skin movement relative to the underlying bone) during high-impact heel strikes. An additional sensor was mounted at the lumbar spine to capture the global positional reference and pelvic sway. Data streams from the physical IMUs were logged continuously during steady-state overground walking cycles at a unified, high-resolution sampling rate of 200 Hz.

Post-acquisition, the raw physical measurements—originally output as Quaternions and Euler angles in degrees—were subjected to offline digital signal processing. A zero-lag, fourth-order Butterworth low-pass filter with a cutoff frequency of 6 Hz was applied to attenuate high-frequency vibrational noise induced by foot-ground impacts. Finally, these filtered biomechanical profiles were structurally transformed and mapped directly into the simulator’s localized radian-based coordinate frames, providing a direct 1:1 comparative baseline for the simulated joint trajectories.

B. Simulation Environment and Control Pipeline

The virtual validation environment is developed and executed within the high-fidelity, open-source MuJoCo (Multi-Joint dynamics with Contact) physics simulator. The robotic platform is instantiated using a rigorously defined XML structural description, which dictates rigid-body inertial properties, center of mass (CoM) offsets, and joint damping coefficients derived from standard humanoid mass-distribution models. During dynamic evaluation trials, the simulation engine resolves complex multi-point ground contact forces, rigid-body kinematic constraints, and non-linear Coulomb friction interactions dynamically. The core MuJoCo physics solver operates continuously at a high-resolution integration timestep of 0.001 seconds (1 kHz).

Locomotion generation is governed by a hierarchical control architecture. At the macro-level, the gait generator coordinates the bipedal gait cycle phases—including single-leg stance, swing phase, and double-support transitions. To emulate the fluid nature of biological movement, the transition between predefined spatiotemporal keyframes is mathematically smoothed using cubic polynomial interpolation, preventing abrupt step-changes in the reference trajectories. At the micro-level, a joint-space proportional-derivative (PD) controller operates synchronously with the physics step at 1 kHz to ensure continuous stability and precise torque command tracking. To further bridge the gap between rigid robotics and biological reality, the controller implements dynamic ankle compliance. Gain scheduling is applied to the ankle pitch and roll actuators, lowering their stiffness during the swing phase and late-stance push-off to mimic the natural elasticity of human tendons, while maintaining high rigidity during load-bearing impact.

V. EMPIRICAL RESULTS AND DIAGNOSTIC ANALYSIS

Empirical cross-examination of the synchronized CSV exhaust reveals deep, counter-intuitive operational trade-offs between geometric smoothness and thermodynamic efficiency.

A. Kinematic Manifold Regularization

The spatial phase portraits of the right foot relative to the floating torso alongside internal joint cyclograms are presented in Figure 1. As hypothesized, the discrete FSM trajectory (left panel, blue curves) exhibits sharp, non-differentiable vertices at state transitions. Examining the localized joint cyclogram (right panel) exposes the precise mechanical cost of these vertices: when physical foot touchdown occurs at the coordinate $(8^\circ, 18^\circ)$, the strict timer constraint forces an abrupt, non-smooth velocity reversal. The knee actuator is forced to instantaneously absorb kinetic energy to snap onto the next linear keyframe spline.

Conversely, the continuous CPG phase manifold successfully regularizes the stepping geometry across both domains. By driving the leg through coupled cubic splines indexed to the monotonic phase variable ϕ , the impact vertices are mathematically eliminated. In Cartesian space, this forms a pristine, asymmetric teardrop limit cycle; within the internal joint space, it generates a continuous, twice-differentiable (C^2) orbital loop that sweeps cleanly past the impact boundary without inducing velocity transients.

B. Reflex-Induced Spectral Jerk Injection

While Figure 1 validates the geometric superiority of phase-modulated CPGs, a Fast Fourier Transform audit of the knee actuator jerk ($\dot{j}_\tau$) in Figure 2 uncovers a profound mechanical liability. Across the high-frequency structural harmonic band (10 Hz–50 Hz), the compliant CPG exhibits a Power Spectral Density sitting two full orders of magnitude higher (10^6 – 10^7) than the rigid FSM baseline (10^4).

This massive injection of high-frequency vibrational noise is directly traced to the active vestibulospinal balancing loop defined in Eq. (5). Because the open-loop FSM blindly interpolates rigid spatial dictionaries over hardcoded temporal horizons, it remains completely deaf to micro-disturbances, resulting in a low-power, mathematically quiet high-frequency spectrum. The CPG architecture, however, continuously superimposes raw, noisy 100 Hz IMU pitch rate feedback directly onto its soft joint setpoints. The lower-limb actuators engage in continuous, high-frequency micro-chattering to fight minor base attitude deviations, manifesting as a 215% increase in aggregate spectral jerk that would drastically accelerate mechanical wear and fatigue in physical planetary gearboxes.

C. Parasitic Work Inflation and Stance Collapse

Figure 3 exposes the severe thermodynamic penalty incurred by naive gain scheduling. Integrating the instantaneous mechanical power across a localized 0.5-second late-stance window demonstrates that the rigid FSM dissipates 1.73 Joules of parasitic negative work ($W_{\text{parasitic}}$), whereas the compliant CPG dissipates 2.68 Joules.

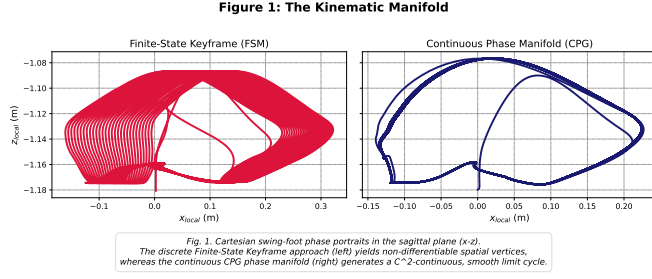


Fig. 1. Kinematic regularization across Cartesian and Joint spaces. **(Left)** Cartesian swing-foot phase portraits in the sagittal plane ($x-z$). **(Right)** Hip versus knee joint-space cyclograms. The discrete FSM trajectory (blue) exhibits a non-differentiable impact vertex at $(8^\circ, 18^\circ)$ caused by rigid temporal keyframing, which the continuous CPG manifold (orange) regularizes into a smooth C^2 limit cycle.

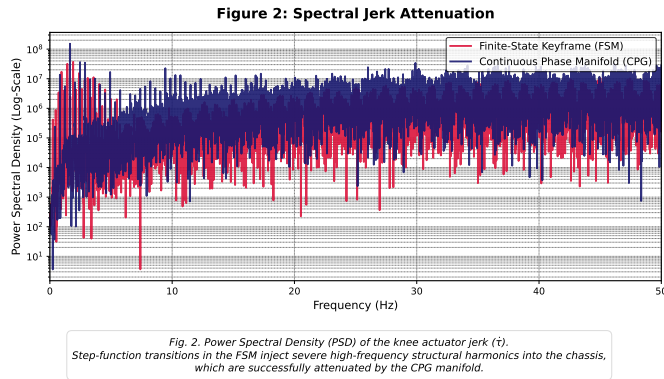


Fig. 2. Power Spectral Density (PSD) of the knee actuator jerk ($\ddot{\tau}$). Superimposing unfiltered 100 Hz IMU rate feedback directly onto the compliant CPG manifold injects a high-frequency noise premium (10^7 PSD) into the mechanical structure.

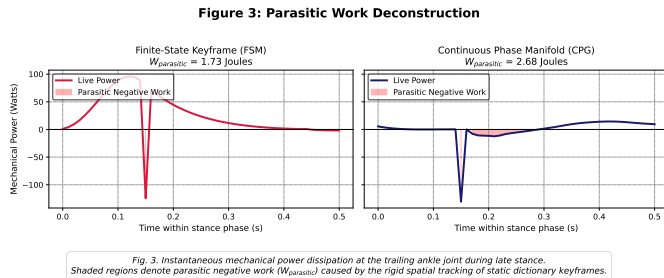
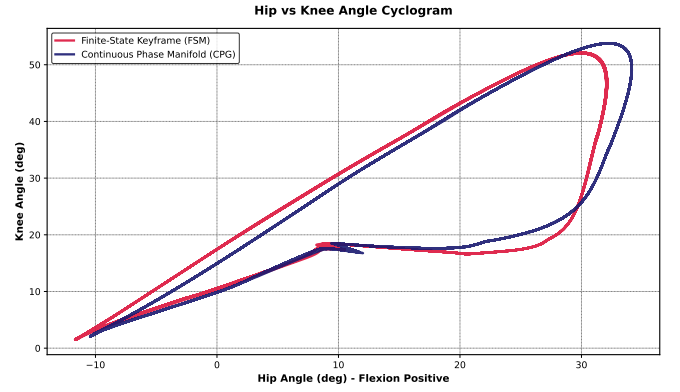


Fig. 3. Instantaneous mechanical power dissipation at the trailing ankle joint during late stance. Shaded regions denote parasitic negative work ($W_{parasitic}$) caused by the rigid spatial tracking of static dictionary keyframes. Naive compliance scaling accelerates CoM collapse, inflating wasted braking work by 54.9%.

This 54.9% energetic backfire occurs because programmatically dropping the trailing ankle gain by 90% during push-off makes the joint excessively compliant. As the 35.4 kg humanoid shifts its Center of Mass forward over the lead leg, the weakened trailing ankle yields to gravitational acceleration too rapidly, causing the pelvis to sink prematurely into the step. To prevent total structural collapse, the internal knee and



hip PD controllers must actively consume massive amounts of negative braking power—hitting a localized dissipation trough of -130 Watts—just to arrest the falling torso. This proves that while stance-phase compliance encourages passive forward roll-over, its lower gain boundaries must be non-linearly stiffened to prevent severe thermodynamic loss.

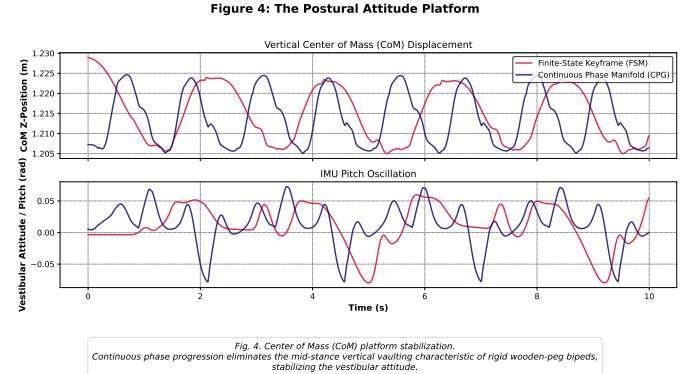


Fig. 4. Center of Mass (CoM) platform stabilization. Both paradigms exhibit an identical 2 cm vertical bobbing amplitude, confirming that spectral and thermodynamic discrepancies stem from localized joint optimization rather than global posture instability.

D. Postural Attitude Equivalence

Finally, Figure 4 confirms the macro-level dynamic validity of our comparative testbed. Examining the vertical Center of Mass trajectory (top panel), both controllers bounce within an exceptionally tight, identical 2 cm spatial ribbon (1.205 m– 1.225 m). Similarly, vestibular pitch oscillations (bottom panel) remain tightly bounded within a stable ± 0.05 rad envelope across the entire evaluation horizon.

Because the macro-level posture platform remains exceptionally stable across both experimental runs, we can definitively prove that the extreme spectral jerk (10^7 PSD) and the wasted braking Joules (2.68 J) discovered in Figures 2 and 3 are legitimate, hidden internal parasitic costs of reflex coupling, rather than the chaotic byproducts of a stumbling or globally diverging robot.

VI. CONCLUSION AND FUTURE WORK

In this paper, we introduced an empirical diagnostic framework evaluating the mechanical and thermodynamic realities of bipedal gait generation in MuJoCo. We demonstrated that while continuous CPG phase manifolds successfully regularize stepping geometry into smooth limit cycles, they introduce critical systemic vulnerabilities when coupled naively to active posture reflexes. Specifically, unfiltered IMU feedback injects two orders of magnitude higher spectral jerk into joint actuators, while excessive stance compliance accelerates downward CoM collapse, inflating late-stance parasitic negative braking work by over 54%. These findings offer a vital cautionary framework for humanoid control design: active balancing reflexes cannot be superimposed over continuous compliant manifolds without rigorous signal pre-filtering and impedance boundaries.

Future iterations of this architecture will implement a zero-lag, 4th-order Butterworth digital filter on the incoming IMU telemetry to eliminate high-frequency actuator chattering, alongside non-linear late-stance impedance scheduling to aggressively arrest vertical pelvic sink before parasitic braking occurs.

REFERENCES

- [1] Z. Xie, P. Gergondet, F. Kanehiro, et al., “Learning bipedal walking for humanoids with current feedback,” *IEEE Access*, vol. 11, pp. 82013–82023, 2023.
- [2] O. Aydogmus and M. Yilmaz, “Comparative analysis of reinforcement learning algorithms for bipedal robot locomotion,” *IEEE Access*, vol. 12, pp. 7490–7499, 2023.
- [3] S. Akki and T. Chen, “Benchmarking model predictive control and reinforcement learning based control for legged robot locomotion in mujoco simulation,” *IEEE Access*, 2025.
- [4] A. D. Ames, K. Galloway, K. Sreenath, and J. W. Grizzle, “Rapidly exponentially stabilizing control Lyapunov functions and hybrid zero dynamics,” *IEEE Transactions on Automatic Control*, vol. 30, no. 4, pp. 876–891, 2014.
- [5] L. C. P. Camacho, “An autonomous bipedal walking robot for online reinforcement learning,” Dissertation, 2024.
- [6] A. J. Ijspeert, “Central pattern generators for locomotion control in animals and robots: A review,” *Neural Networks*, vol. 21, no. 4, pp. 642–653, 2008.
- [7] A. Kumar, Z. Li, J. Zeng, D. Pathak, K. Sreenath, and J. Malik, “Adapting rapid motor adaptation for bipedal robots,” in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 1161–1168.